

TeProM 5

English IV Seminar 6

Academic and Professional Writing 5: Lab Reports

1 Overview

Laboratory reports are written for several reasons. One reason is to communicate the laboratory work to management. In such situations, management often bases company decisions on the results of the report. Another reason to write laboratory reports is to record the work so that the work will not have to be done in the future. This handout presents one commonly used organization for laboratory reports:

- Abstract
- Introduction
- Procedures
- Results and Discussion
- Conclusions
- References
- Appendices

You should not assume, however, that this organization will serve all your laboratory reports. In other words, one organization does not "fit" all experiments or tasks. Rather, you should pay attention to the organization requested by your instructor, who has chosen an organization that best serves your experiments.

2 Abstract

The abstract presents a synopsis of the experiment. The following guidelines for preparing an abstract are from the American Institute of Aeronautics and Astronautics (AIAA). Note that although your instructor may define the term "abstract" differently, these guidelines still give you a sense of the stylistic issues that distinguish abstracts, such as whether to include numerical data:

The abstract should be written concisely in normal rather than highly abbreviated English. The author should assume that the reader has some knowledge of the subject but has not read the paper. Thus, the abstract should be intelligible and complete in itself; particularly it should not cite figures, tables, or sections of the paper. The opening sentence or two should, in general, indicate the subjects dealt with in the paper and should state the objectives of the investigation. It is also desirable to describe the treatment by one or more such terms as brief, exhaustive, theoretical, experimental, and so forth.

The body of the abstract should indicate newly observed facts and the conclusions of the experiment or argument discussed in the paper. It should contain new numerical data presented in the paper if space permits; otherwise, attention should be drawn to the nature of such data. In the case of experimental results, the abstract should indicate the methods used in obtaining them; for new methods the basic principle, range of operation, and degree of accuracy should be given. The abstract should be typed as one paragraph. Its optimum length will vary somewhat with the nature and extent of the paper, but it should not exceed 200 words.

3 Introduction

The "Introduction" of a laboratory report identifies the experiment to be undertaken, the objectives of the experiment, the importance of the experiment, and the overall background for understanding the experiment. The objectives of the experiment are important to state because these objectives are usually analyzed in the discussion or conclusion to determine whether the experiment succeeded. The background often includes theoretical predictions for what the results should be.

4 Procedures

The "Procedures," often called "Materials and Methods," discusses how the experiment was carried out. Documenting the materials and procedures of your laboratory experiment is important not only so that others can repeat your results but also so that you can replicate the work later, if the need arises. For further information on the "Procedures" section, see the previous handout.

5 Results and Discussion

The heart of a laboratory report is the presentation of the results and the discussion of those results. In some reports, "Results" and "Discussion" appear as separate sections. The decision to separate or combine these sections depends upon your experiment and the purpose of your laboratory report. Therefore, pay attention to what your laboratory instructor requests. Also, use your judgment. For instance, it is a good idea to combine these sections when the discussion of your first result is needed to understand your second result. On the other hand, it is probably better to separate these sections when you want to discuss the results as a whole after all results are reported.

In discussing the results, you should not only analyze the results, but also discuss the implications of those results. Moreover, pay attention to the errors that occurred in the experiment, both where they originated and what their significance is for interpreting the reliability of your conclusions. One important way to present numerical results is to show them in graphs.

6 Conclusions

A "Conclusion" section often appears in longer laboratory reports, and in reports where the "Results and Discussion" sections are combined. The "Conclusion" section, like the separate "Discussion" section in shorter reports, discusses the results in the context of the entire experiment. Usually, the objectives mentioned in the "Introduction" are examined to determine whether the experiment succeeded. If the objectives were not met, you should analyze why the results were not as predicted.

7 References

The "References" section gives a list of the texts, whether paper or electronic, that were used to prepare for, carry out, and write the lab exercise.

8 Appendices

Appendices are often included in a laboratory report. An appendix in a laboratory report typically presents information that is too detailed to be put in the report's text. For example, if you have a long table, you might place this tabular information in an appendix and include a graph of the data in the report's text. Another type of appendix that can appear in laboratory

source: adapted from <http://www.writing.eng.vt.edu/workbooks/laboratory.html>

reports is one that gives supplementary information not directly concern the experiment's objectives.

If the appendix is "formal," it should contain a beginning, middle, and ending. For example, if the appendix contains tables of test data, the appendix should not only contain the tabular data, but also formally introduce those tables, discuss why they have been included, and explain the unusual aspects that might confuse the reader. Because of time restrictions, your instructor might allow you to include "informal" appendices with calculations and supplemental information. For such "informal" situations, having a clear beginning, middle, and ending is not necessary. However, you should still give the appendix a title, place a heading on each table, place a caption beneath each figure, and insert comments necessary for reader understanding.

8 Deciding on a Structure

In an educational context, the structure for your lab report will often be prescribed by the course instructor. If the structure is not prescribed, choose a structure that is suited to the nature and length of the report. However, even the shortest report should at least include an introduction, a description of the materials and procedure used in the experiment or task, the results of the experiment or task, and some discussion of the results. The sample lab report accompanying this handout gives just one example of the structure that can be used for a lab report (source: <http://www.physics.mun.ca/~cdeacon/labs/simonfraser.pdf>).